

Bessel Functions

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- 5 Bridge Proofs for Scattering Applications
 - Deriving the Cylindrical Far-Field Factor
 - Exact Spherical Outgoing Functions and Modal Limits

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Introduction

Bessel functions arise naturally when separation of variables is applied in cylindrical coordinates, for example in circular waveguides, circular membranes, and diffraction by circular apertures. Their dimensionless argument will be denoted by x . Unless otherwise stated, the limiting formulas below assume $x > 0$ and fixed real order $\nu \geq 0$. Bessel's differential equation is

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0. \quad (\text{B.1.1})$$

On any interval that excludes the singular point $x = 0$, two linearly independent solutions are the Bessel functions of the first and second kinds, $J_\nu(x)$ and $Y_\nu(x)$.

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Bessel Functions of the First and Second Kind

For $\nu \notin \{-1, -2, \dots\}$, the Bessel function of the first kind is represented, with a consistent branch for x^ν , by

$$J_\nu(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(n + \nu + 1)} \left(\frac{x}{2}\right)^{2n+\nu}. \quad (\text{B.2.1})$$

Termwise differentiation verifies the definition directly. If a_n denotes the n th term, then

$$x^2 a_n'' + x a_n' - \nu^2 a_n = [(2n + \nu)^2 - \nu^2] a_n = 4n(n + \nu) a_n,$$

whereas, for $n \geq 1$,

$$x^2 a_{n-1} = -4n(n + \nu) a_n.$$

Thus the $x^2 a_{n-1}$ term cancels the differential contribution of a_n term by term, leaving $x^2 J_\nu'' + x J_\nu' + (x^2 - \nu^2) J_\nu = 0$.

The Gamma function appearing here is defined for $\Re z > 0$ by

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt,$$

Bessel Functions of the First and Second Kind (cont.)

and integration by parts gives

$$\begin{aligned}\Gamma(z+1) &= \int_0^\infty t^z e^{-t} dt \\ &= \left[-t^z e^{-t}\right]_0^\infty + z \int_0^\infty t^{z-1} e^{-t} dt = z\Gamma(z),\end{aligned}$$

where the boundary term vanishes for $\Re z > 0$. Hence $\Gamma(n+1) = n!$ for integers $n \geq 0$. For noninteger ν , the Bessel function of the second kind, or Neumann function, is defined by

$$Y_\nu(x) = \frac{J_\nu(x) \cos(\nu\pi) - J_{-\nu}(x)}{\sin(\nu\pi)}. \quad (\text{B.2.2})$$

For an integer order m , both numerator and denominator vanish, and the definition is continued by the limit

$$Y_m(x) = \lim_{\nu \rightarrow m} \frac{J_\nu(x) \cos(\nu\pi) - J_{-\nu}(x)}{\sin(\nu\pi)}. \quad (\text{B.2.3})$$

Bessel Functions of the First and Second Kind (cont.)

The nonzero Wronskian derived below proves that J_ν and Y_ν are linearly independent. Therefore, on an interval with $x > 0$, the general solution is

$$y(x) = c_1 J_\nu(x) + c_2 Y_\nu(x). \quad (\text{B.2.4})$$

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Small- and Large-Argument Behavior

For fixed $\nu > 0$, the leading $n = 0$ term of (B.2.1) gives

$$J_\nu(x) \sim \frac{1}{\Gamma(\nu+1)} \left(\frac{x}{2}\right)^\nu, \quad x \rightarrow 0^+. \quad (\text{B.3.1})$$

In particular, for the nonnegative orders used in regular cylindrical-field problems, $J_0(0) = 1$ and $J_\nu(x) \rightarrow 0$ as $x \rightarrow 0^+$ when $\nu > 0$. The often-repeated statement that J_ν is finite at the origin requires this order restriction: for a negative noninteger order it generally diverges. Negative integer orders satisfy $J_{-n}(x) = (-1)^n J_n(x)$. For positive noninteger ν , $J_{-\nu}$ dominates J_ν as $x \rightarrow 0^+$. Inserting its leading term into (B.2.2) and then using $\Gamma(\nu)\Gamma(1-\nu) = \pi/\sin(\pi\nu)$ gives the intermediate steps

$$Y_\nu(x) \sim -\frac{1}{\sin(\pi\nu)\Gamma(1-\nu)} \left(\frac{x}{2}\right)^{-\nu},$$
$$\frac{1}{\sin(\pi\nu)\Gamma(1-\nu)} = \frac{\Gamma(\nu)}{\pi}.$$

Continuation to positive integer order gives the same leading result:

$$Y_\nu(x) \sim -\frac{\Gamma(\nu)}{\pi} \left(\frac{2}{x}\right)^\nu, \quad x \rightarrow 0^+, \quad \nu > 0. \quad (\text{B.3.2})$$

Small- and Large-Argument Behavior (cont.)

Let γ_E denote Euler's constant. The order-zero case is different and must not be obtained by substituting $\nu = 0$ into (B.3.2):

$$\begin{aligned} J_0(x) &= 1 - \frac{x^2}{4} + O(x^4), \\ Y_0(x) &= \frac{2}{\pi} \left[\ln\left(\frac{x}{2}\right) + \gamma_E \right] + O(x^2 \ln x). \end{aligned} \tag{B.3.3}$$

Thus Y_ν is singular at the origin for every nonnegative real order and is excluded when a physical field must remain bounded at the cylinder axis.

For fixed ν and real $x \rightarrow +\infty$, define

$$\chi_\nu = x - \frac{\nu\pi}{2} - \frac{\pi}{4}.$$

The leading large-argument asymptotic forms are

$$J_\nu(x) \sim \sqrt{\frac{2}{\pi x}} \cos \chi_\nu, \tag{B.3.4a}$$

$$Y_\nu(x) \sim \sqrt{\frac{2}{\pi x}} \sin \chi_\nu. \tag{B.3.4b}$$

Small- and Large-Argument Behavior (cont.)

When $x = k\rho$ with real $k > 0$, the factor $x^{-1/2}$ is the $\rho^{-1/2}$ amplitude spreading of a cylindrical wave. Spherical waves instead have amplitude proportional to r^{-1} ; the comparison is about radial distance, not about an arbitrary dimensionless argument. The Hankel functions are defined with correlated upper and lower signs:

$$\begin{aligned}H_{\nu}^{(1,2)}(x) &= J_{\nu}(x) \pm iY_{\nu}(x), \\H_{\nu}^{(1)}(x) &= J_{\nu}(x) + iY_{\nu}(x), \\H_{\nu}^{(2)}(x) &= J_{\nu}(x) - iY_{\nu}(x).\end{aligned}\tag{B.3.5}$$

For $\nu > 0$, the dominant small-argument terms are therefore

$$H_{\nu}^{(1,2)}(x) \sim \mp i \frac{\Gamma(\nu)}{\pi} \left(\frac{2}{x}\right)^{\nu}, \quad x \rightarrow 0^{+}.\tag{B.3.6}$$

Keeping only this leading asymptotic is deliberate: for noninteger order, the subleading terms of Y_{ν} mean that simply appending the leading term of J_{ν} does not give a generally valid two-term expansion. At order zero,

$$H_0^{(1,2)}(x) = 1 \pm \frac{2i}{\pi} \left[\ln\left(\frac{x}{2}\right) + \gamma_E \right] + O(x^2 \ln x).$$

Small- and Large-Argument Behavior (cont.)

Combining (B.3.4a) and (B.3.4b) with $\cos \chi \pm i \sin \chi = e^{\pm i\chi}$ gives

$$H_{\nu}^{(1,2)}(x) \sim \sqrt{\frac{2}{\pi x}} e^{\pm i\chi_{\nu}}, \quad x \rightarrow +\infty. \quad (\text{B.3.7})$$

The propagation labels depend on the time convention. With $e^{i\omega t}$ and $x = k\rho$,

$$\Re\{e^{i\omega t} H_{\nu}^{(1)}(k\rho)\} \propto \cos(\omega t + k\rho - \text{constant}), \quad \frac{d\rho}{dt} = -\frac{\omega}{k},$$

so $H_{\nu}^{(1)}$ is inward-propagating, while

$$\Re\{e^{i\omega t} H_{\nu}^{(2)}(k\rho)\} \propto \cos(\omega t - k\rho - \text{constant}), \quad \frac{d\rho}{dt} = +\frac{\omega}{k},$$

so $H_{\nu}^{(2)}$ is outward-propagating. The roles reverse if the $e^{-i\omega t}$ convention is used.

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Derivation of Important Relations

The following identities are stated with their hypotheses and derivation steps so that signs and normalizations can be checked directly.

Recurrence and derivative relations. For any cylinder function $Z_\nu \in \{J_\nu, Y_\nu, H_\nu^{(1)}, H_\nu^{(2)}\}$, differentiation of its series or connection formula gives

$$\frac{d}{dx} [x^\nu Z_\nu(x)] = x^\nu Z_{\nu-1}(x), \quad (\text{B.4.1a})$$

$$\frac{d}{dx} [x^{-\nu} Z_\nu(x)] = -x^{-\nu} Z_{\nu+1}(x). \quad (\text{B.4.1b})$$

Applying the product rule and dividing by $x^{\pm\nu}$ gives

$$Z'_\nu + \frac{\nu}{x} Z_\nu = Z_{\nu-1}, \quad Z'_\nu - \frac{\nu}{x} Z_\nu = -Z_{\nu+1}.$$

Adding and subtracting these equations yields

$$Z_{\nu-1}(x) + Z_{\nu+1}(x) = \frac{2\nu}{x} Z_\nu(x), \quad (\text{B.4.2a})$$

$$Z_{\nu-1}(x) - Z_{\nu+1}(x) = 2Z'_\nu(x). \quad (\text{B.4.2b})$$

Derivation of Important Relations (cont.)

Wronskians. After division by x^2 , Bessel's equation has the form

$$y'' + \frac{1}{x}y' + \left(1 - \frac{\nu^2}{x^2}\right)y = 0.$$

For two solutions y_1, y_2 , define $W = y_1y_2' - y_1'y_2$. Direct differentiation and substitution of the differential equation give

$$W' = y_1y_2'' - y_1''y_2 = -\frac{1}{x}W \implies W = \frac{C}{x}.$$

Using the small- x forms (B.3.1)–(B.3.2) for $\nu > 0$ fixes C :

$$J_\nu Y_\nu' - J_\nu' Y_\nu \sim \frac{2\nu}{x} \frac{1}{2^\nu \Gamma(\nu + 1)} \frac{2^\nu \Gamma(\nu)}{\pi} = \frac{2}{\pi x}.$$

Continuation in order includes $\nu = 0$, so

$$J_\nu(x)Y_\nu'(x) - J_\nu'(x)Y_\nu(x) = \frac{2}{\pi x}, \quad (\text{B.4.3a})$$

$$J_\nu(x)H_\nu^{(1,2)'}(x) - J_\nu'(x)H_\nu^{(1,2)}(x) = \pm \frac{2i}{\pi x}. \quad (\text{B.4.3b})$$

Derivation of Important Relations (cont.)

The second identity follows immediately by inserting $H_\nu^{(1,2)} = J_\nu \pm iY_\nu$; the $J_\nu - J_\nu$ Wronskian is zero.

Generating function and integer-order integrals. For $t \neq 0$,

$$e^{\frac{x}{2}(t-1/t)} = \sum_{n=-\infty}^{\infty} J_n(x) t^n. \quad (\text{B.4.4})$$

To verify the coefficient, expand the two exponentials:

$$e^{xt/2} e^{-x/(2t)} = \sum_{p=0}^{\infty} \frac{(xt/2)^p}{p!} \sum_{q=0}^{\infty} \frac{(-x/(2t))^q}{q!}.$$

For $n \geq 0$, the coefficient of t^n has $p = q + n$ and is

$$\sum_{q=0}^{\infty} \frac{(-1)^q}{q!(q+n)!} \left(\frac{x}{2}\right)^{2q+n} = J_n(x);$$

negative n follows from $J_{-n} = (-1)^n J_n$.

Derivation of Important Relations (cont.)

Setting $t = e^{i\theta}$ in (B.4.4) gives

$$e^{ix \sin \theta} = \sum_{n=-\infty}^{\infty} J_n(x) e^{in\theta}.$$

Fourier coefficient extraction therefore yields, for integer n ,

$$J_n(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{i(x \sin \theta - n\theta)} d\theta \quad (\text{B.4.5a})$$

$$= \frac{1}{\pi} \int_0^{\pi} \cos(n\theta - x \sin \theta) d\theta \quad (\text{B.4.5b})$$

$$= \frac{1}{2\pi i^n} \int_0^{2\pi} e^{i(x \cos \theta + n\theta)} d\theta. \quad (\text{B.4.5c})$$

The middle formula is not valid for arbitrary noninteger ν without an additional integral term; restricting it to $n \in \mathbb{Z}$ is essential.

Derivation of Important Relations (cont.)

Orthogonality and normalization at zeros. Let $a > 0$, let $\alpha_{\nu m} > 0$ be the m th positive zero of J_ν for $m = 1, 2, \dots$, and define

$$R_m(\rho) = J_\nu\left(\frac{\alpha_{\nu m}\rho}{a}\right), \quad 0 \leq \rho \leq a.$$

The radial Bessel equation is in Sturm–Liouville form,

$$\frac{d}{d\rho} \left(\rho \frac{dR_m}{d\rho} \right) + \left(\frac{\alpha_{\nu m}^2}{a^2} \rho - \frac{\nu^2}{\rho} \right) R_m = 0.$$

Multiply the m equation by R_n , multiply the n equation by R_m , subtract, and integrate. The result is

$$\left(\frac{\alpha_{\nu m}^2 - \alpha_{\nu n}^2}{a^2} \right) \int_0^a \rho R_m R_n d\rho = \left[\rho (R_m R'_n - R_n R'_m) \right]_0^a.$$

The endpoint at a vanishes because $R_m(a) = R_n(a) = 0$, and regularity makes the endpoint at 0 vanish. Hence the integral is zero for $m \neq n$.

Derivation of Important Relations (cont.)

For $m = n$, set $u = \alpha_{\nu n}\rho/a$ and use the directly differentiable identity

$$\frac{d}{du} \left\{ \frac{u^2}{2} [J_{\nu}^2(u) - J_{\nu-1}(u)J_{\nu+1}(u)] \right\} = uJ_{\nu}^2(u).$$

It gives

$$\begin{aligned} \int_0^a \rho R_n^2 d\rho &= \frac{a^2}{\alpha_{\nu n}^2} \int_0^{\alpha_{\nu n}} u J_{\nu}^2(u) du \\ &= \frac{a^2}{2} [J_{\nu}^2(\alpha_{\nu n}) - J_{\nu-1}(\alpha_{\nu n})J_{\nu+1}(\alpha_{\nu n})] \\ &= \frac{a^2}{2} J_{\nu+1}^2(\alpha_{\nu n}), \end{aligned}$$

because $J_{\nu}(\alpha_{\nu n}) = 0$ and the recurrence relation gives $J_{\nu-1}(\alpha_{\nu n}) = -J_{\nu+1}(\alpha_{\nu n})$.
Combining both cases (with δ_{mn} the Kronecker delta),

$$\int_0^a J_{\nu}\left(\frac{\alpha_{\nu m}\rho}{a}\right) J_{\nu}\left(\frac{\alpha_{\nu n}\rho}{a}\right) \rho d\rho = \frac{a^2}{2} J_{\nu+1}^2(\alpha_{\nu n}) \delta_{mn}. \quad (\text{B.4.6})$$

Derivation of Important Relations (cont.)

Circular-waveguide TE modes instead use positive zeros $\alpha'_{\nu n}$ of J'_ν . The same Sturm–Liouville subtraction proves orthogonality: at $\rho = a$ both radial derivatives vanish, while regularity again makes the $\rho = 0$ endpoint term vanish. At a derivative zero, the recurrence equations give $J_{\nu-1}(\alpha') = J_{\nu+1}(\alpha') = (\nu/\alpha')J_\nu(\alpha')$, so the normalization is

$$\begin{aligned} \int_0^a J_\nu\left(\frac{\alpha'_{\nu m}\rho}{a}\right) J_\nu\left(\frac{\alpha'_{\nu n}\rho}{a}\right) \rho d\rho \\ = \frac{a^2}{2} \left(1 - \frac{\nu^2}{(\alpha'_{\nu n})^2}\right) J_\nu^2(\alpha'_{\nu n}) \delta_{mn}. \end{aligned} \tag{B.4.7}$$

Only positive derivative zeros are included; the formal zero of J'_0 at the origin is not a nontrivial TE₀₀ waveguide mode.

Derivation of Important Relations

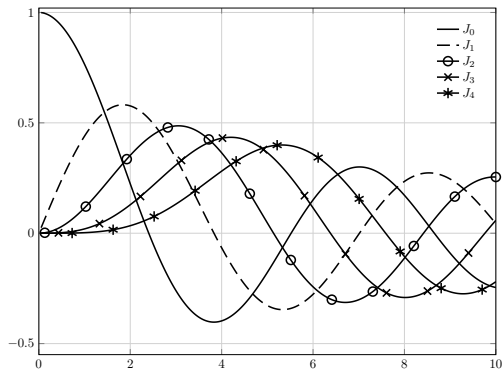


Figure B.1: Bessel functions of the first kind for $\nu = 0, 1, 2, 3, 4$.

Derivation of Important Relations

Derivation of Important Relations

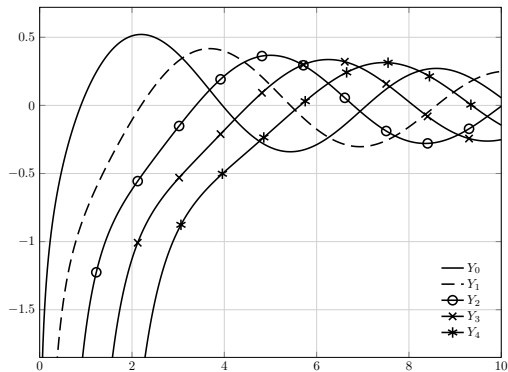


Figure B.2: Bessel functions of the second kind for $\nu = 0, 1, 2, 3, 4$.

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Bridge Proofs for Scattering Applications

This section supplies the derivations on which Chapter 4 depends and which are not carried out there: the Frobenius construction behind (B.2.1), a proof of the generating function (B.4.4), the fixed-order meaning of the large-argument forms, the spherical functions with the plane-wave integral representation, the Riccati–Bessel Wronskian, and the small-argument Rayleigh series. The Gamma reflection identity and beta–gamma relations used below are explicit special-function prerequisites [?].

Frobenius construction and the integer-order limit. Put $B = \sum_{q \geq 0} c_q z^{q+s}$ with $c_0 \neq 0$ in $z^2 B'' + z B' + (z^2 - \nu^2) B = 0$. Since $z^2 B = \sum_q c_{q-2} z^{q+s}$, the lowest power gives the indicial equation $(s^2 - \nu^2) c_0 = 0$, hence $s = \pm \nu$, and the remaining powers give

$$[(q+s)^2 - \nu^2] c_q + c_{q-2} = 0, \quad c_{-1} \equiv 0.$$

Choose $s = \nu \geq 0$. The odd coefficients vanish, and

$$c_{2p} = \frac{(-1)^p \Gamma(\nu + 1) c_0}{4^p p! \Gamma(p + \nu + 1)}.$$

The conventional normalization $c_0 = [2^\nu \Gamma(\nu + 1)]^{-1}$ yields exactly (B.2.1). For integer n , the terms of the J_{-n} series with negative powers vanish at the zeros of

Bridge Proofs for Scattering Applications (cont.)

$1/\Gamma$ —equivalently, at the poles of Γ —shifting the surviving series into $J_{-n} = (-1)^n J_n$. Both numerator and denominator of (B.2.2) then vanish at $\nu = n$, and differentiating with respect to order (L'Hôpital's rule) gives the explicit limit form

$$Y_n(z) = \frac{1}{\pi} \left[\partial_\nu J_\nu(z)|_{\nu=n} + (-1)^n \partial_\nu J_\nu(z)|_{\nu=-n} \right],$$

which exhibits Y_n as a second independent solution; independence itself follows from the nonzero Wronskian below. The same order-derivative produces the logarithmic constant in (B.3.3). At $\nu = 0$ the limit formula gives $Y_0 = (2/\pi) \partial_\nu J_\nu|_{\nu=0}$, and differentiating the series (B.2.1) termwise in ν gives, with $\psi = \Gamma'/\Gamma$ the digamma function and H_n the n th harmonic number,

$$\begin{aligned} Y_0(x) &= \frac{2}{\pi} \sum_{n \geq 0} \frac{(-1)^n}{(n!)^2} \left(\frac{x}{2}\right)^{2n} \left[\ln \frac{x}{2} - \psi(n+1) \right] \\ &= \frac{2}{\pi} \left[\ln \frac{x}{2} + \gamma_E \right] J_0(x) - \frac{2}{\pi} \sum_{n \geq 1} \frac{(-1)^n H_n}{(n!)^2} \left(\frac{x}{2}\right)^{2n}, \end{aligned}$$

using $\psi(n+1) = H_n - \gamma_E$; the leading term is (B.3.3).

Bridge Proofs for Scattering Applications (cont.)

Proof of the generating function. Expand both exponentials in $\exp[(z/2)(t - t^{-1})] = \exp(zt/2) \exp[-z/(2t)]$:

$$\sum_{p,q \geq 0} \frac{(-1)^q (z/2)^{p+q}}{p! q!} t^{p-q}.$$

For $n \geq 0$, the coefficient of t^n has $p = q + n$ and equals the J_n series (B.2.1); negative coefficients follow from $J_{-n} = (-1)^n J_n$. Hence the sum is $\sum_{n \in \mathbb{Z}} J_n(z) t^n$, proving (B.4.4). For fixed finite z and $|t| = 1$, absolute convergence of the exponential product permits the regrouping. Setting $t = -ie^{i\phi}$ gives $t - t^{-1} = -2i \cos \phi$, and therefore

$$e^{-iz \cos \phi} = \sum_{n \in \mathbb{Z}} i^{-n} J_n(z) e^{in\phi},$$

from which Fourier orthogonality yields the cylindrical plane-wave expansion used in Chapter 4 without any unproved identity.

Fixed-order meaning of the large-argument forms. The general-order complex-sector forms (B.3.4a)–(B.3.7) are reference asymptotic expansions [?]. For fixed ν they apply as $|z| \rightarrow \infty$: the J_ν , Y_ν sectors are $|\arg z| \leq \pi - \delta$, the $H_\nu^{(1)}$ sector is

Bridge Proofs for Scattering Applications (cont.)

$-\pi + \delta \leq \arg z \leq 2\pi - \delta$, and the $H_\nu^{(2)}$ sector is $-2\pi + \delta \leq \arg z \leq \pi - \delta$, with fixed $\delta > 0$ and the corresponding analytically continued branches. Their error constants depend on order. For J_ν, Y_ν one must not interpret an envelope error as a relative error at a zero of the leading sine or cosine. The integer-order positive-real-axis factors used for cylindrical waves are derived in Section B.5.1. The spherical outgoing functions and their derivatives are obtained from an exact finite formula in Section B.5.2.

In particular, $kr \gg 1$ alone is not a uniform error criterion for arbitrarily large n . Section B.5.2 gives two separate controls for Chapter 4: a majorant for the angular modal tail, and a majorant for the exact radial functions uniform in $r \geq b > a$. The first omitted *majorant term*, rather than the first actual omitted modal term, controls the tail. These bounds are for fixed admissible material parameters and size, not uniformly across resonant or singular material limits.

Spherical Bessel functions and the plane-wave integral. For integer $n \geq 0$ define, with the principal square root,

$$j_n(z) = \sqrt{\frac{\pi}{2z}} J_{n+1/2}(z), \quad y_n(z) = \sqrt{\frac{\pi}{2z}} Y_{n+1/2}(z),$$
$$h_n^{(1,2)}(z) = j_n(z) \pm i y_n(z),$$

Bridge Proofs for Scattering Applications (cont.)

and the Riccati–Bessel functions $\hat{j}_n(z) = zj_n(z)$, $\hat{y}_n(z) = zy_n(z)$, $\hat{h}_n^{(1,2)}(z) = zh_n^{(1,2)}(z)$, using the hat notation of Chapter 4. Let

$$K_n(z) = \int_{-1}^1 e^{-izu} (1 - u^2)^n du.$$

Expanding the exponential, the odd terms integrate to zero, and the even moments are beta integrals:

$$K_n(z) = \sum_{p \geq 0} \frac{(-1)^p z^{2p}}{(2p)!} B\left(p + \frac{1}{2}, n + 1\right).$$

With $B(p + \frac{1}{2}, n + 1) = \Gamma(p + \frac{1}{2}) n! / \Gamma(p + n + \frac{3}{2})$ and the duplication consequence $\Gamma(p + \frac{1}{2}) / (2p)! = \sqrt{\pi} / (4^p p!)$,

$$j_n(z) = \frac{z^n}{2^{n+1} n!} K_n(z). \tag{B.5.1}$$

Bridge Proofs for Scattering Applications (cont.)

The interchange of series and integral is justified by uniform convergence on the finite interval for fixed finite z . Rodrigues' formula and n integrations by parts (all endpoint terms vanish since $(u^2 - 1)^n$ has order- n zeros at ± 1) then give

$$\int_{-1}^1 e^{-izu} P_n(u) du = \frac{(-i)^n z^n}{2^n n!} K_n(z) = 2(-i)^n j_n(z),$$

from which projection with the Legendre norm yields the spherical plane-wave coefficient $(2n+1)(-i)^n$ with no circular premise.

Riccati–Bessel Wronskian. For $\psi_n(z) = \sqrt{\pi z/2} J_{n+1/2}(z)$ and $\xi_n(z) = \sqrt{\pi z/2} H_{n+1/2}^{(2)}(z)$, the derivatives of the common square-root factor cancel from the Wronskian, so with $W(J_\nu, H_\nu^{(2)}) = -2i/(\pi z)$,

$$W(\psi_n, \xi_n) = \frac{\pi z}{2} \left(-\frac{2i}{\pi z} \right) = -i. \quad (\text{B.5.2})$$

With the equation ordering of the Chapter 4 elimination, this is the $-i$ Cramer numerator of the internal Mie coefficients c_n and d_n . Neither elimination divides by ψ_n

Bridge Proofs for Scattering Applications (cont.)

or ψ'_n , so the formulas remain usable at individual interior Bessel zeros whenever the full modal determinant is nonzero.

Rayleigh series and general-multipole order. From the half-integer series or their recurrences,

$$\psi_1(z) = \frac{\sin z}{z} - \cos z = \frac{z^2}{3} - \frac{z^4}{30} + O(z^6),$$

$$\xi_1(z) = -e^{-iz}(1 - i/z) = \frac{i}{z} + \frac{iz}{2} + \frac{z^2}{3} - \frac{iz^3}{8} + O(z^4),$$

with derivatives $\psi'_1 = 2z/3 - 2z^3/15 + O(z^5)$ and $\xi'_1 = -i/z^2 + i/2 + 2z/3 + O(z^2)$, including all signs. For general fixed $n \geq 1$,

$$\psi_n(z) \sim \frac{z^{n+1}}{(2n+1)!!}, \quad \xi_n(z) \sim i(2n-1)!! z^{-n}.$$

Bridge Proofs for Scattering Applications (cont.)

Inserting these and their derivatives into the Chapter 4 determinant quotients gives, at fixed contrast with $x = ka$ and $\kappa = k_d/k$, numerator order x^{2n+1} against a nonzero quasistatic denominator:

$$\begin{aligned} a_n &= \frac{i(n+1)}{(2n+1)!!(2n-1)!!} \frac{\epsilon_d - \epsilon}{n\epsilon_d + (n+1)\epsilon} x^{2n+1} + O(x^{2n+3}), \\ b_n &= \frac{i(n+1)}{(2n+1)!!(2n-1)!!} \frac{\mu_d - \mu}{n\mu_d + (n+1)\mu} x^{2n+1} + O(x^{2n+3}). \end{aligned} \tag{B.5.3}$$

These are equalities with remainder rather than asymptotic equivalences: the displayed leading coefficient can vanish, since the numerator series contains only the odd powers $x^{2n+1}, x^{2n+3}, \dots$ and the denominator is a nonzero constant plus $O(x^2)$. When $\epsilon_d = \epsilon$ (respectively $\mu_d = \mu$), the contrast factor is zero and the formula then asserts only $a_n = O(x^{2n+3})$ (respectively for b_n). For $n = 1$ and nonzero contrast this reproduces the $2ix^3(\epsilon_d - \epsilon)/[3(\epsilon_d + 2\epsilon)]$ dipole. The nonzero-denominator hypotheses stated in Chapter 4 are precisely what permit the order comparison, and the next denominator term is $O(x^2)$, exhibiting the dynamic-detuning scale.

Deriving the Cylindrical Far-Field Factor

The integer-order positive-real-axis asymptotics used in Chapter 4 can be obtained from an integral, including their phase and normalization. For $\text{Im } z < 0$ define

$$I(z) = \int_0^\infty e^{-iz \cosh t} dt.$$

The integral and its derivatives converge absolutely. Differentiating and integrating a total derivative gives

$$z^2 I'' + zI' + z^2 I = -iz \int_0^\infty \frac{d}{dt} (\sinh t e^{-iz \cosh t}) dt = 0.$$

The boundary term is zero at both ends. To identify which order-zero Bessel solution this represents, put $z = -is$, $s > 0$, and $u = e^t$:

$$I(-is) = \int_1^\infty e^{-s(u+u^{-1})/2} \frac{du}{u}.$$

Deriving the Cylindrical Far-Field Factor (cont.)

Its difference from $E_1(s/2) = \int_1^\infty e^{-su/2} du/u$ is $O(s)$, since $|1 - e^{-s/(2u)}| \leq s/(2u)$. Splitting the defining integral of E_1 at 1 gives

$$\begin{aligned}\lim_{a \rightarrow 0^+} [E_1(a) + \ln a] &= \int_0^1 \frac{e^{-v} - 1}{v} dv + \int_1^\infty \frac{e^{-v}}{v} dv \\ &= \Gamma'(1) = -\gamma_E.\end{aligned}$$

For the second equality, integrate

$$\Gamma'(1) = \int_0^\infty e^{-v} \ln v dv$$

by parts with a lower cutoff, and then take its limit. Hence

$I(-is) = -\ln(s/2) - \gamma_E + o(1)$. The already derived small-argument forms of J_0 , Y_0 , with $\ln(-is) = \ln s - i\pi/2$, give the identical logarithmic and constant terms for $(\pi/2i)[J_0(-is) - iY_0(-is)]$. A difference of order-zero solutions is a linear combination of J_0 and Y_0 ; zero logarithmic coefficient and zero constant force both coefficients to vanish. Analytic continuation within the lower half-plane therefore proves

$$H_0^{(2)}(z) = \frac{2i}{\pi} I(z).$$

Deriving the Cylindrical Far-Field Factor (cont.)

Approach $z = x > 0$ from below, retaining this limiting convergence factor. With $w = 2 \sinh(t/2)$ the integral becomes

$$I(x) = e^{-ix} \int_0^\infty \frac{e^{-ixw^2/2}}{\sqrt{1 + w^2/4}} dw.$$

The stationary endpoint is $w = 0$. Its amplitude is $1 + O(w^2)$. The Gaussian integral follows by first using a positive real damping factor and then taking its limit:

$$\lim_{\delta \downarrow 0} \int_0^\infty e^{-(\delta+i)xw^2/2} dw = e^{-i\pi/4} \sqrt{\frac{\pi}{2x}}.$$

The real Gaussian normalization is obtained by squaring the integral and changing to polar coordinates; its continuation to $\text{Re}(\delta + i) > 0$ fixes the square-root branch. For the remainder, insert a smooth cutoff equal to one near zero. The endpoint amplitude minus one starts with w^2 ; scaling $w = x^{-1/2}v$ makes its leading contribution $O(x^{-3/2})$. Each

Deriving the Cylindrical Far-Field Factor (cont.)

further even Taylor term is smaller by x^{-1} . On the cutoff complement the phase derivative is nonzero, and repeated use of

$$e^{-ixw^2/2} = -\frac{1}{ixw} \frac{d}{dw} e^{-ixw^2/2}$$

bounds the remainder after integration by parts. The limiting damping factor controls the endpoint at infinity. This gives

$$H_0^{(2)}(x) = \sqrt{\frac{2}{\pi x}} e^{-i(x-\pi/4)} [1 + O(x^{-1})].$$

The same cutoff argument applied before removing the damping factor gives differentiated expansions; it is not necessary to differentiate an unspecified big- O term. In particular, $(H_0^{(2)})' = -i\sqrt{2/(\pi x)} e^{-i(x-\pi/4)} [1 + O(x^{-1})]$. The recurrence already derived in this appendix, $H_{n+1}^{(2)} = nH_n^{(2)}/x - (H_n^{(2)})'$, now gives by induction

$$H_n^{(2)}(x) = \sqrt{\frac{2}{\pi x}} e^{-i(x-n\pi/2-\pi/4)} [1 + O_n(x^{-1})]$$

Deriving the Cylindrical Far-Field Factor (cont.)

for each fixed nonnegative integer n . For real x , $H_n^{(1)} = (H_n^{(2)})^*$; taking their half-sum and difference also gives the corresponding J_n , Y_n expansions. The errors for J_n , Y_n are additive relative to the oscillation envelope, not relative to a sine or cosine that may be zero. This proof covers the integer-order real-axis wave interpretation; it is not a proof of every general-order complex-sector expansion recorded elsewhere in this appendix.

Exact Spherical Outgoing Functions and Modal Limits

Put $\psi_n(z) = zj_n(z)$ and $\xi_n(z) = zh_n^{(2)}(z)$. The half-integer series and Gamma recurrence give $\psi_0 = \sin z$, $zy_0 = -\cos z$, and hence $\xi_0 = ie^{-iz}$ and $\xi_1 = -e^{-iz}(1 - i/z)$. The recurrence $\xi_{n+1} = (2n+1)\xi_n/z - \xi_{n-1}$ yields the exact finite formula

$$\xi_n(z) = i^{n+1} e^{-iz} Q_n(z), \quad Q_n(z) = \sum_{m=0}^n A_{nm} \left(-\frac{i}{z}\right)^m,$$
$$A_{nm} = \frac{(n+m)!}{2^m m! (n-m)!}.$$

$A_{n+1,m} = A_{n-1,m} + (2n+1)A_{n,m-1}$, with out-of-range coefficients set to zero, is precisely the coefficient identity obtained from that recurrence, and the two initial functions fix the solution. Since $A_{n0} = 1$ and $A_{n1} = n(n+1)/2$, at fixed n

$$\xi_n(z) = i^{n+1} e^{-iz} \left[1 - \frac{in(n+1)}{2z} + O_n(z^{-2}) \right].$$

Derivatives follow by differentiating the finite polynomial Q_n , and the formula applies for complex $z \neq 0$, fixing the outgoing spherical phase without importing a half-integer asymptotic theorem.

Exact Spherical Outgoing Functions and Modal Limits (cont.)

Large order at fixed material and size. For the sphere calculation fix $x = ka > 0$, $y = k_d a \neq 0$, and finite nonzero $\epsilon, \mu, \epsilon_d, \mu_d$, with $y^2/x^2 = \epsilon_d \mu_d / (\epsilon \mu)$; the exterior medium is lossless, and exact zeros of the modal determinants used below are excluded. These are fixed-parameter assertions, not bounds uniform across resonances or singular material limits. Write $d_n = (2n - 1)!!$, $D_n = (2n + 1)!!$, and

$$\psi_n(z) = \frac{z^{n+1}}{D_n} F_n(z), \quad zy_n(z) = -d_n z^{-n} V_n(z).$$

The regular series gives

$$F_n(z) = \sum_{p=0}^{\infty} \frac{(-1)^p z^{2p}}{2^p p! \prod_{j=1}^p (2n + 2j + 1)},$$

Exact Spherical Outgoing Functions and Modal Limits (cont.)

and inserting $z^{-n} \sum_p v_p z^{2p}$ into $u'' + [1 - n(n+1)/z^2]u = 0$ gives $v_p = v_{p-1}/[2p(2n+1-2p)]$; its leading normalization, fixed by the half-integer definition, yields the convergent series

$$V_n(z) = \sum_{p=0}^{\infty} \frac{z^{2p}}{2^p p! \prod_{j=1}^p (2n+1-2j)},$$

whose denominator factors never vanish (empty products are one). For $n \geq 1$ the inequality $p|2n+1-2p| \geq n$ holds for every $p \geq 1$ (the left side is a concave quadratic on $1 \leq p \leq n$ and increases from $n+1$ for $p \geq n+1$), so the absolute successive-term ratios in F_n, V_n are bounded by $|z|^2/[2(2n+3)]$ and $|z|^2/(2n)$. For either series $W = F_n$ or $W = V_n$, let $q < 1$ be its stated absolute successive-term ratio bound. Since the constant term is one,

$$|W - 1| \leq \sum_{p=1}^{\infty} q^p = \frac{q}{1-q}, \quad |zW'| \leq \sum_{p=1}^{\infty} 2pq^p = \frac{2q}{(1-q)^2}.$$

Exact Spherical Outgoing Functions and Modal Limits (cont.)

The second estimate bounds the scaled ordinary derivative. To divide by W , require $q < 1/2$; then

$$|W| \geq \frac{1-2q}{1-q} > 0, \quad \left| \frac{zW'}{W} \right| \leq \frac{2q}{(1-q)(1-2q)}.$$

At fixed nonzero z , both ratio bounds are $O_z(n^{-1})$ and eventually satisfy $q < 1/2$. Consequently, at fixed nonzero z ,

$$F_n = 1 + O_z(n^{-1}), \quad V_n = 1 + O_z(n^{-1}), \\ zF'_n, zV'_n = O_z(n^{-1}).$$

Set $L_\psi(z) = z\psi'_n(z)/\psi_n(z)$ and $L_\xi(z) = z\xi'_n(z)/\xi_n(z)$. Since $\xi_n = \psi_n - ixy_n$ and $w_{n+1}/w_n = x^2/[(2n+1)(2n+3)] \rightarrow 0$ for $w_n = x^{2n+1}/(d_n D_n)$, the regular contribution

Exact Spherical Outgoing Functions and Modal Limits (cont.)

to ξ_n and its derivative is smaller than the irregular one, and the preceding estimates give, for sufficiently large n ,

$$L_\psi(z) = n + 1 + O_z(n^{-1}), \quad L_\xi(x) = -n + O_x(n^{-1}),$$
$$|\psi_n(x)/\xi_n(x)| \leq C_x w_n.$$

The exact interface equations, rather than independently chosen square roots, reduce the Mie quotients at sufficiently large n to

$$a_n = \frac{\psi_n(x)}{\xi_n(x)} \frac{L_\psi(y) - \epsilon_r L_\psi(x)}{L_\psi(y) - \epsilon_r L_\xi(x)},$$
$$b_n = \frac{\psi_n(x)}{\xi_n(x)} \frac{L_\psi(y) - \mu_r L_\psi(x)}{L_\psi(y) - \mu_r L_\xi(x)},$$

where $\epsilon_r = \epsilon_d/\epsilon$ and $\mu_r = \mu_d/\mu$. These logarithmic forms are used only past the threshold where their factors are nonzero; the original determinant formulas must still be used at individual Bessel zeros. For either fixed contrast s , the new denominator

$$L_\psi(y) - sL_\xi(x) = n(1 + s) + 1 + O(n^{-1})$$

Exact Spherical Outgoing Functions and Modal Limits (cont.)

grows linearly if $s \neq -1$ and tends to one if $s = -1$, while the numerator is at most $O(n)$. Hence

$$|a_n| + |b_n| \leq C(n+1)w_n \quad \text{for all sufficiently large } n. \quad (\text{B.5.4})$$

The threshold and constant depend on the fixed parameters; for a PEC sphere, the direct formulas $a_n = \psi'_n/\xi'_n$ and $b_n = \psi_n/\xi_n$ give the same bound without taking a material limit.

An explicit angular tail majorant. A conservative bound suffices; a sharp angular estimate is unnecessary. Expanding Rodrigues' polynomial gives

$$P_n(u) = 2^{-n} \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j (2n-2j)!}{j!(n-j)!(n-2j)!} u^{n-2j},$$

whose absolute coefficient at index j is $2^{-n} \binom{2n-2j}{n-j} \binom{n-j}{j} \leq 4^n 8^{-j}$. Thus for $|u| \leq 1$ the absolute-coefficient sum is at most $(8/7)4^n$, so $|P'_n(u)| \leq \frac{8}{7}n4^n$ and $|P''_n(u)| \leq \frac{8}{7}n^2 4^n$. With $u = \cos \theta$, the phase-free angular functions are $\pi_n = P'_n(u)$ and $\tau_n = uP'_n(u) - (1-u^2)P''_n(u)$, including their continuous pole limits. Combining their

Exact Spherical Outgoing Functions and Modal Limits (cont.)

bounds with (B.5.4) and $(2n+1)/[n(n+1)]$ bounds each far-field amplitude summand by

$$M_n = C(n+1)^4 4^n w_n,$$
$$\frac{M_{n+1}}{M_n} = \frac{4x^2}{(2n+1)(2n+3)} \left(\frac{n+2}{n+1}\right)^4 \rightarrow 0.$$

Choose N beyond the coefficient threshold and sufficiently large that this ratio is at most a fixed $q < 1$ for every $n \geq N+1$. If t_n is the actual summand, then

$$\left| \sum_{n>N} t_n \right| \leq \sum_{n>N} M_n \leq \frac{M_{N+1}}{1-q}. \quad (\text{B.5.5})$$

The numerator is the first omitted *majorant*, not the magnitude of the first actual omitted modal term, which may vanish through its angular factor or cancellation while later terms do not. The exponential angular bound is deliberately loose; factorial decay still makes the majorant summable.

Exact Spherical Outgoing Functions and Modal Limits (cont.)

Justifying the far-field limit of the exact radial sums. The angular-series estimate alone does not justify applying a fixed-order radial asymptotic inside an infinite series. To close that step, fix any $b > a$ and set $z = kr \geq z_0 = kb > 0$. From the finite outgoing polynomial define $T_{nm} = A_{nm}z_0^{-m}$; its last term is $T_{nn} = d_n z_0^{-n}$. For $0 \leq j \leq n$,

$$\frac{T_{n,n-j}}{T_{nn}} = \frac{(2z_0)^j}{j!} \prod_{\ell=0}^{j-1} \frac{n-\ell}{2n-\ell} \leq \frac{z_0^j}{j!},$$

so, uniformly for real $z \geq z_0$,

$$\begin{aligned} |Q_n(z)| &\leq e^{z_0} d_n z_0^{-n}, & |Q'_n(z)| &\leq (n/z_0) e^{z_0} d_n z_0^{-n}, \\ |Q''_n(z)| &\leq [n(n+1)/z_0^2] e^{z_0} d_n z_0^{-n}. \end{aligned}$$

After multiplying the Mie and angular estimates, the exact scaled potential and field terms and the radial derivatives used in their reconstruction are bounded by a sequence of the form

$$B_n = C(n+1)^8 \frac{x(4x^2/z_0)^n}{(2n+1)!!}.$$

Exact Spherical Outgoing Functions and Modal Limits (cont.)

The power eight is a convenient overestimate of the finitely many polynomial factors involved. Since

$$\frac{B_{n+1}}{B_n} = \left(\frac{n+2}{n+1}\right)^8 \frac{4x^2}{z_0(2n+3)} \rightarrow 0,$$

the exact radial series have a summable majorant independent of $r \geq b$, so dominated convergence permits $r \rightarrow \infty$ to pass through them, including the first radial derivative used in Chapter 4. This proves the termwise far-field rule for each fixed admissible sphere problem. It does not supply a parameter-uniform bound when a determinant tends to zero or when $|k_d a| \rightarrow \infty$; in particular, the PEC formulas are obtained directly from the PEC boundary conditions, without claiming an unproved interchange of an infinite material limit and a modal sum.